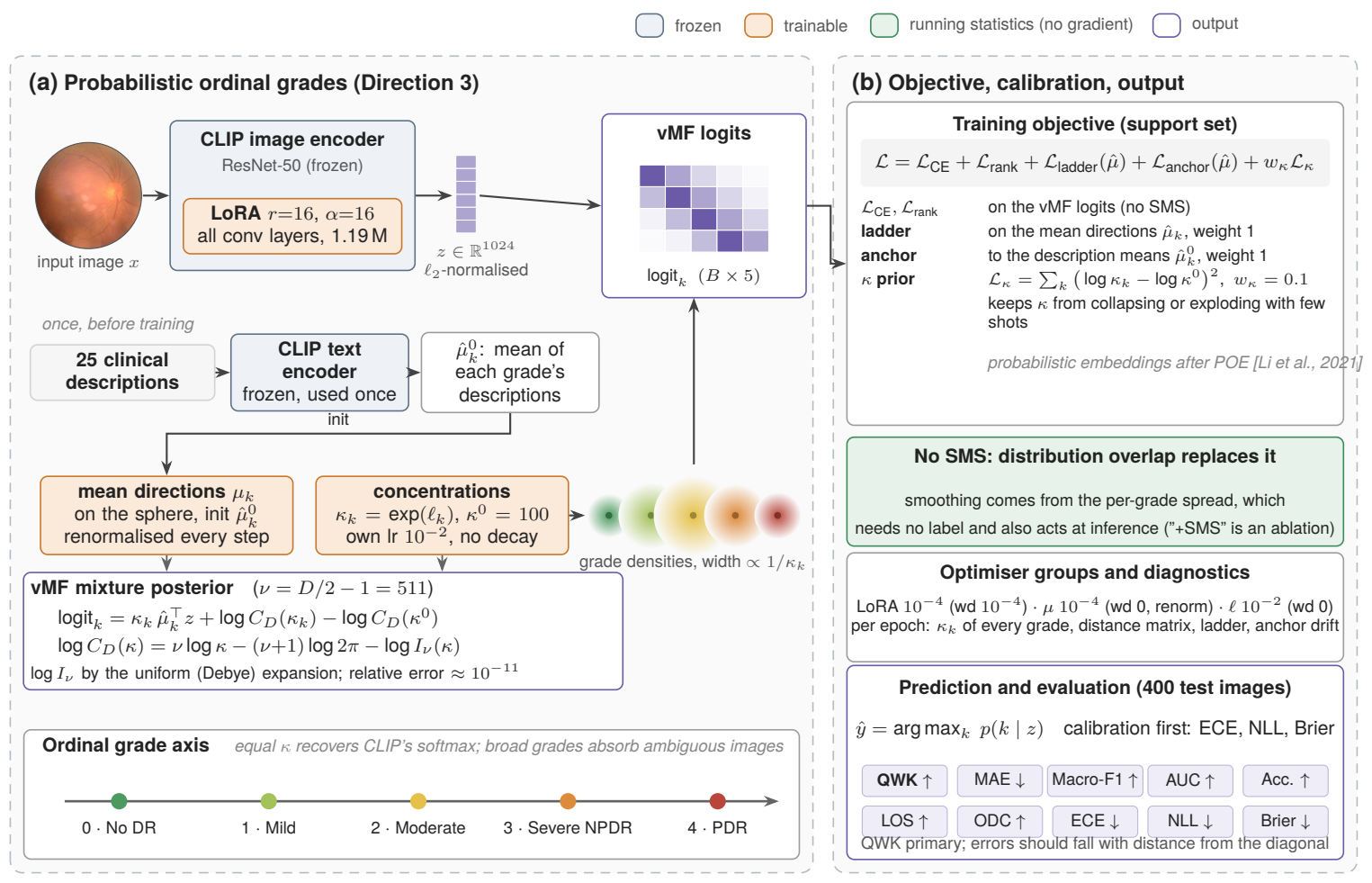
frozen
trainable
running statistics (no gradient)
output
(a) Probabilistic ordinal grades (Direction 3)
input image x
CLIP image encoder
ResNet-50 (frozen)
LoRA r=16, α=16
all conv layers, 1.19 M
$z \in \mathbb{R}^{1024}$
ℓ_2-normalised
vMF logits
logit_k $(B \times 5)$
once, before training
25 clinical descriptions
CLIP text encoder
frozen, used once
init
$\hat{\mu}_k^0$: mean of each grade's descriptions
mean directions μ_k
on the sphere, init $\hat{\mu}_k^0$
renormalised every step
concentrations
$\kappa_k = \exp(\ell_k)$, $\kappa^0 = 100$
own lr 10^{-2}, no decay
grade densities, width $\propto 1/\kappa_k$
vMF mixture posterior $(\nu = D/2 - 1 = 511)$
$\text{logit}_k = \kappa_k \hat{\mu}_k^\top z + \log C_D(\kappa_k) - \log C_D(\kappa^0)$
$\log C_D(\kappa) = \nu \log \kappa - (\nu{+}1) \log 2\pi - \log I_\nu(\kappa)$
$\log I_\nu$ by the uniform (Debye) expansion; relative error $\approx 10^{-11}$
Ordinal grade axis
equal κ recovers CLIP's softmax; broad grades absorb ambiguous images
0 · No DR
1 · Mild
2 · Moderate
3 · Severe NPDR
4 · PDR
(b) Objective, calibration, output
Training objective (support set)
$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{rank}} + \mathcal{L}_{\text{ladder}}(\hat{\mu}) + \mathcal{L}_{\text{anchor}}(\hat{\mu}) + w_\kappa \mathcal{L}_\kappa$
$\mathcal{L}_{\text{CE}}$, $\mathcal{L}_{\text{rank}}$ on the vMF logits (no SMS)
ladder on the mean directions $\hat{\mu}_k$, weight 1
anchor to the description means $\hat{\mu}_k^0$, weight 1
κ prior $\mathcal{L}_\kappa = \sum_k (\log \kappa_k - \log \kappa^0)^2$, $w_\kappa = 0.1$
keeps κ from collapsing or exploding with few shots
probabilistic embeddings after POE [Li et al., 2021]
No SMS: distribution overlap replaces it
smoothing comes from the per-grade spread, which needs no label and also acts at inference ("+SMS" is an ablation)
Optimiser groups and diagnostics
LoRA 10^{-4} (wd 10^{-4}) · μ 10^{-4} (wd 0, renorm) · ℓ 10^{-2} (wd 0)
per epoch: κ_k of every grade, distance matrix, ladder, anchor drift
Prediction and evaluation (400 test images)
$\hat{y} = \arg\max_k \; p(k \mid z)$ calibration first: ECE, NLL, Brier
QWK ↑
MAE ↓
Macro-F1 ↑
AUC ↑
Acc. ↑
LOS ↑
ODC ↑
ECE ↓
NLL ↓
Brier ↓
QWK primary; errors should fall with distance from the diagonal